

Multivariate Analysis

Part 3: More Visualizations!

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Agenda

1. Overtime Polls + `weighted.mean()`
2. Predicting winner with Electoral College Votes

2020 polling

- Back into the data!
 - Download and open [Pres2020_PV.Rds](#)
 - Wrangle to get the popular vote margin, expressed in decimals

```
require(tidyverse)
poll <-
read_rds('https://github.com/jbisbee1/DS1000_F2024/raw/main/data/Pres2020_PV.Rds')

poll <- poll %>%
  mutate(Trump = Trump/100,
         Biden = Biden/100,
         margin = Biden - Trump)
```

Theory

- Previous results indicate that polls which predict greater support for Biden **also** predict greater support for Trump
 - Is this consistent with the theory?
 - Recall that **Biden-biased** polls should underpredict Trump support and overpredict Biden support
 - In the **data**, this would suggest a **negative** relationship
 - But we find a **positive** relationship
- **Inconsistent** with the theory, but raises another puzzle
- Why do polls that underpredict support for Biden also underpredict support for Trump?
 - **Third party bias?** Polls bias against 3rd party candidates
 - **Timing of poll?** Fewer uncertain responses closer to election

Dates and More Dimensions

- To test the [timing of polls theory](#), let's look at how predicted support varies over time

```
poll %>%  
  select(StartDate,EndDate,DaysinField)
```

```
## # A tibble: 528 × 3  
##   StartDate   EndDate   DaysinField  
##   <chr>      <chr>      <dbl>  
## 1 10/31/2020 11/2/2020      3  
## 2 10/31/2020 11/2/2020      3  
## 3 10/29/2020 11/2/2020      5  
## 4 11/1/2020  11/1/2020      1  
## 5 11/1/2020  11/1/2020      1  
## 6 10/30/2020 11/1/2020      3  
## 7 10/31/2020 11/2/2020      3  
## 8 10/30/2020 11/1/2020      3  
## 9 10/29/2020 11/1/2020      4  
## 10 10/29/2020 11/1/2020      4  
## # i 518 more rows
```

Dates

- Need to convert `StartDate` and `EndDate` into `date` class

```
election.day <- as.Date("11/3/2020", "%m/%d/%Y")  
election.day
```

```
## [1] "2020-11-03"
```

```
election.day16 <- as.Date("11/8/2016", "%m/%d/%Y")  
election.day16
```

```
## [1] "2016-11-08"
```

```
election.day - election.day16
```

```
## Time difference of 1456 days
```

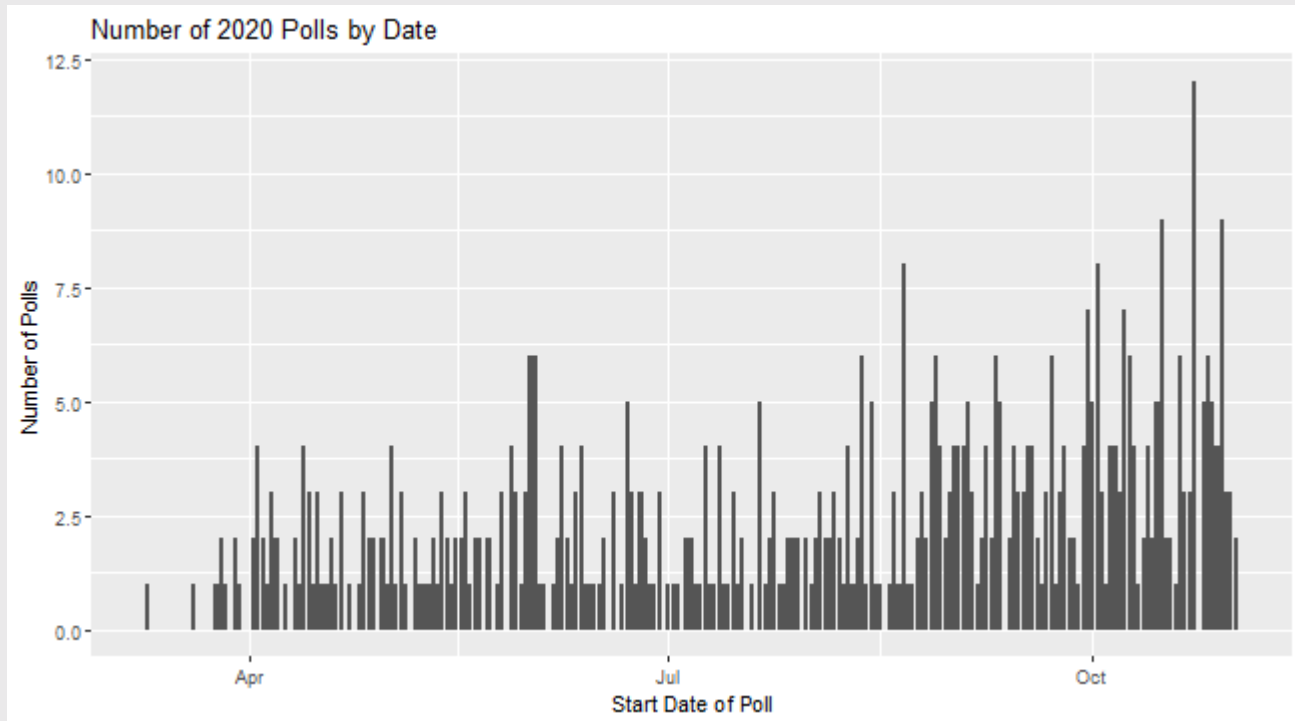
Wrangling

```
toplot <- poll %>%  
  mutate(EndDate = as.Date(EndDate, "%m/%d/%Y"),  
         StartDate = as.Date(StartDate, "%m/%d/%Y"),  
         DaysToED = as.numeric(election.day - EndDate),  
         margin = Biden - Trump)
```

- How many polls were conducted over time?

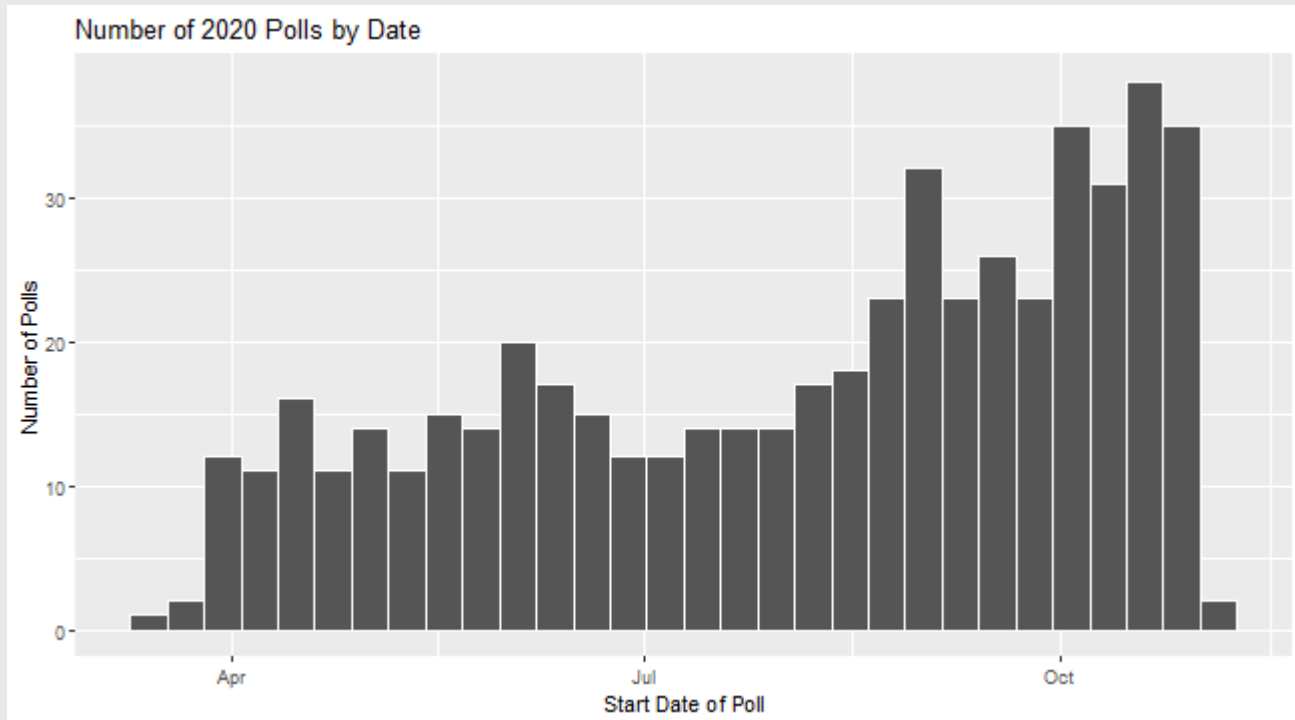
Polls by Date (`geom_bar`)

```
toplot %>%  
  ggplot(aes(x = StartDate)) +  
  geom_bar() + labs(title = 'Number of 2020 Polls by Date', x = 'Start  
Date of Poll', y = 'Number of Polls')
```



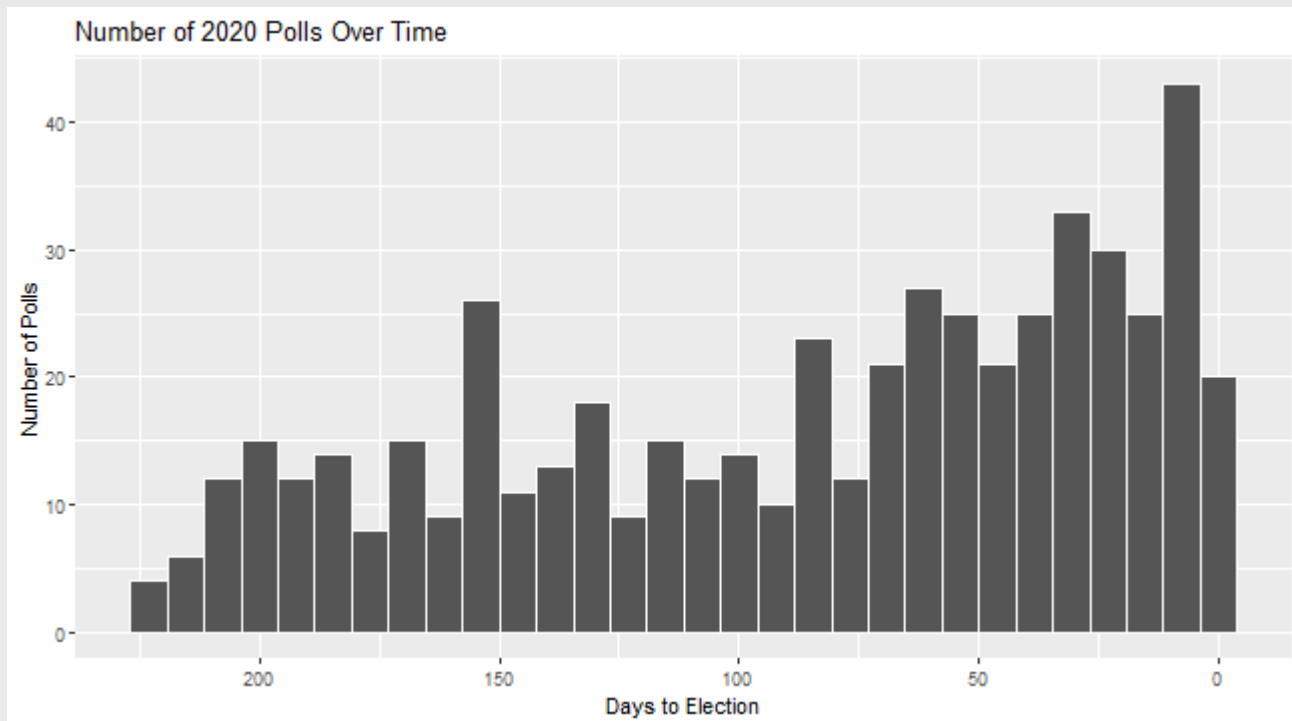
Polls by Date (`geom_histogram`)

```
toplot %>%  
  ggplot(aes(x = StartDate)) +  
  geom_histogram(bins = 30, color = 'white') + labs(title = 'Number of  
2020 Polls by Date', x = 'Start Date of Poll', y = 'Number of Polls')
```



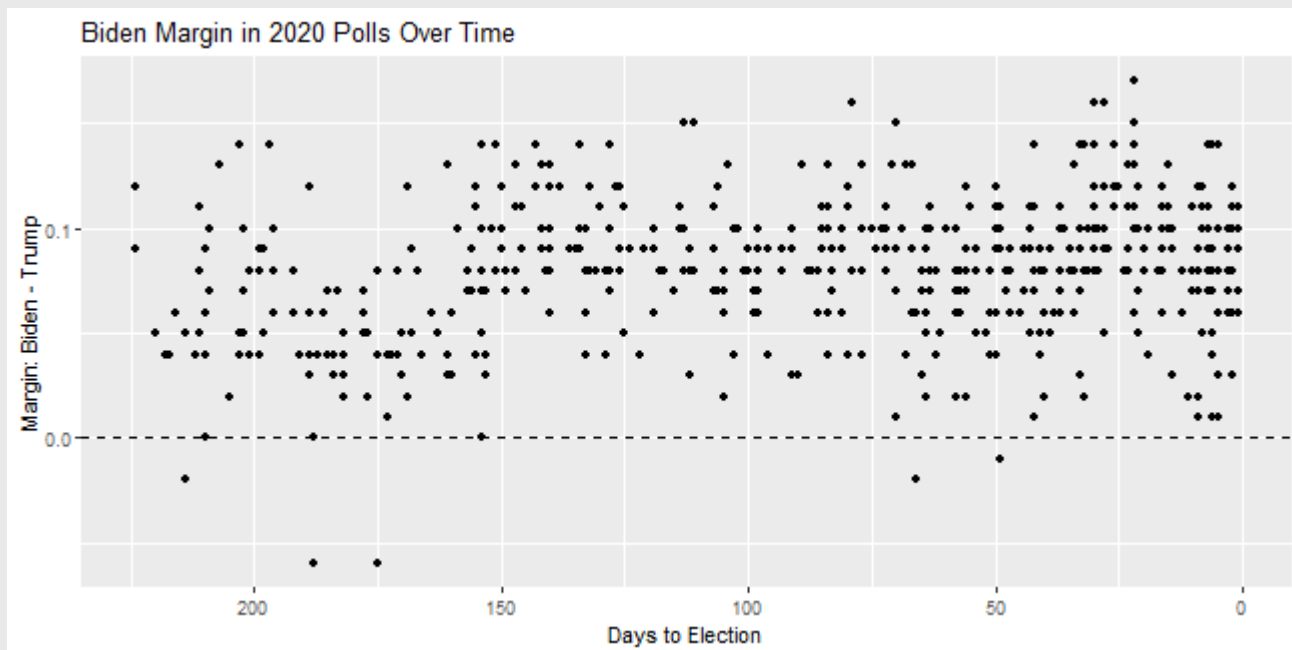
Polls by Time to Election

```
toplot %>%  
  ggplot(aes(x = DaysToED)) +  
  geom_histogram(bins = 30, color = 'white') + labs(title = 'Number of  
2020 Polls Over Time', x = 'Days to Election', y = 'Number of Polls') +  
  scale_x_reverse()
```



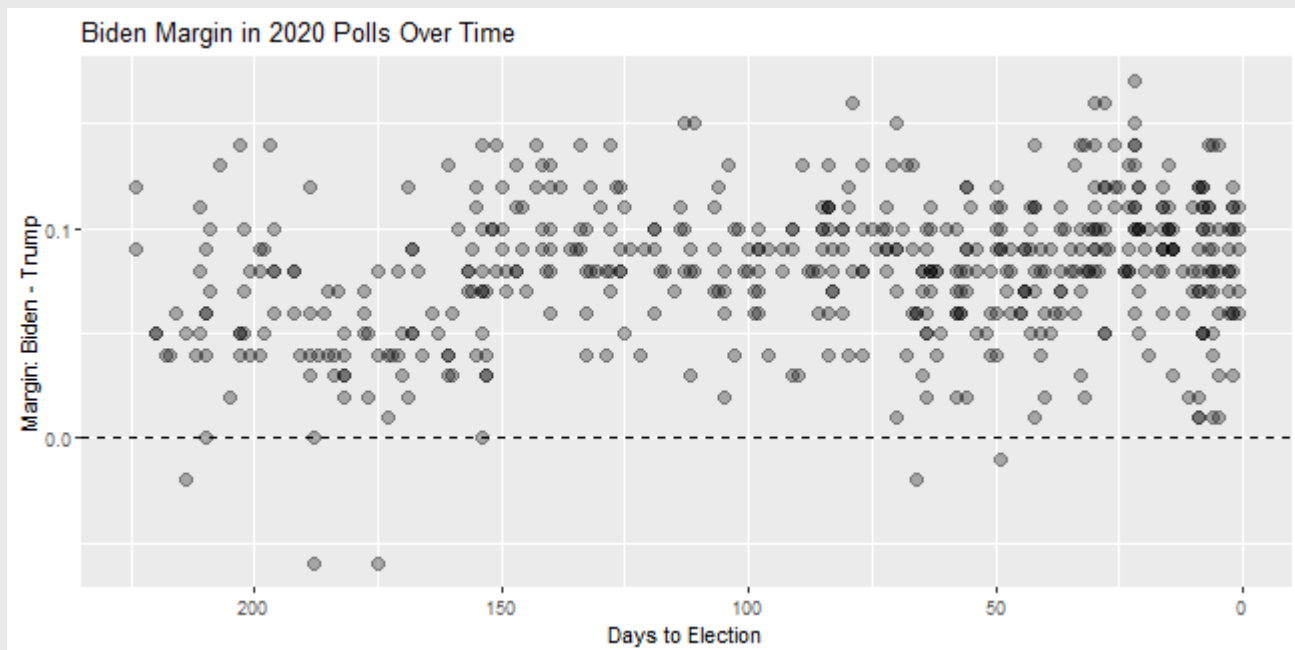
Margin by Time to Election

```
toplot %>%  
  ggplot(aes(x = DaysToED,y = margin)) +  
  geom_point() +  
  labs(title = 'Biden Margin in 2020 Polls Over Time',  
        x = 'Days to Election',y = 'Margin: Biden - Trump') +  
  scale_x_reverse() + geom_hline(yintercept = 0,linetype = 'dashed')
```



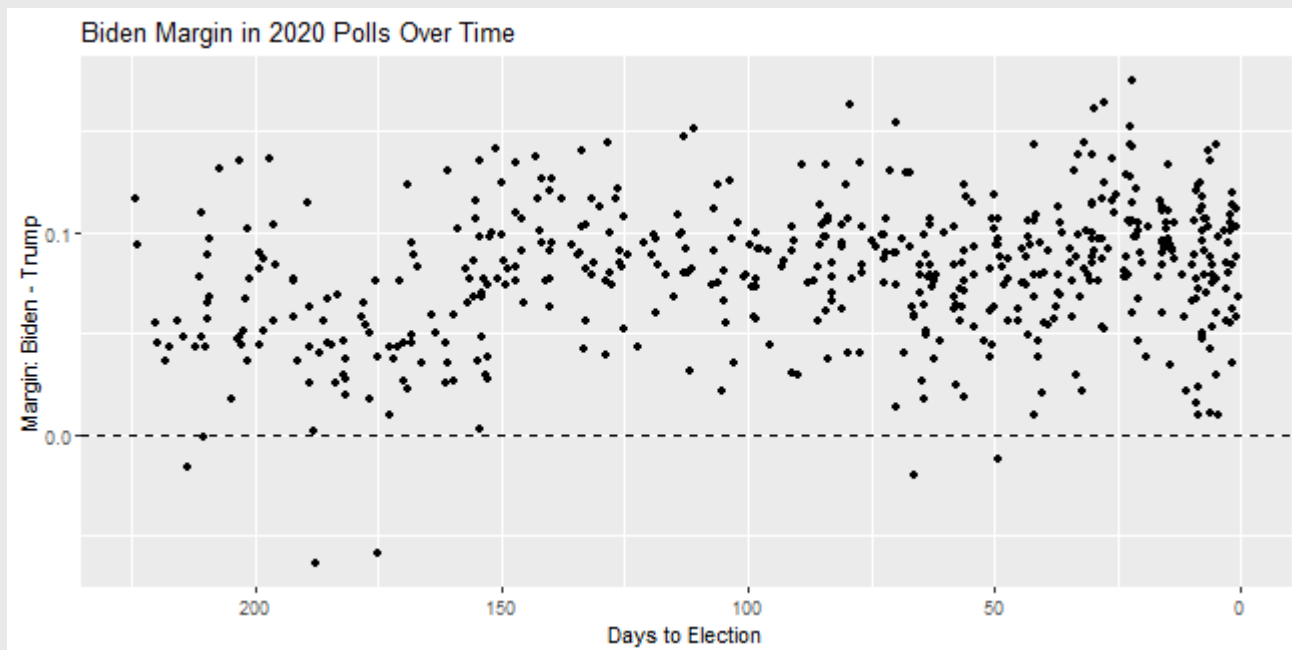
Margin by Time to Election

```
toplot %>%  
  ggplot(aes(x = DaysToED,y = margin)) +  
  geom_point(alpha = .3,size = 3) +  
  labs(title = 'Biden Margin in 2020 Polls Over Time',  
       x = 'Days to Election',y = 'Margin: Biden - Trump') +  
  scale_x_reverse() + geom_hline(yintercept = 0,linetype = 'dashed')
```



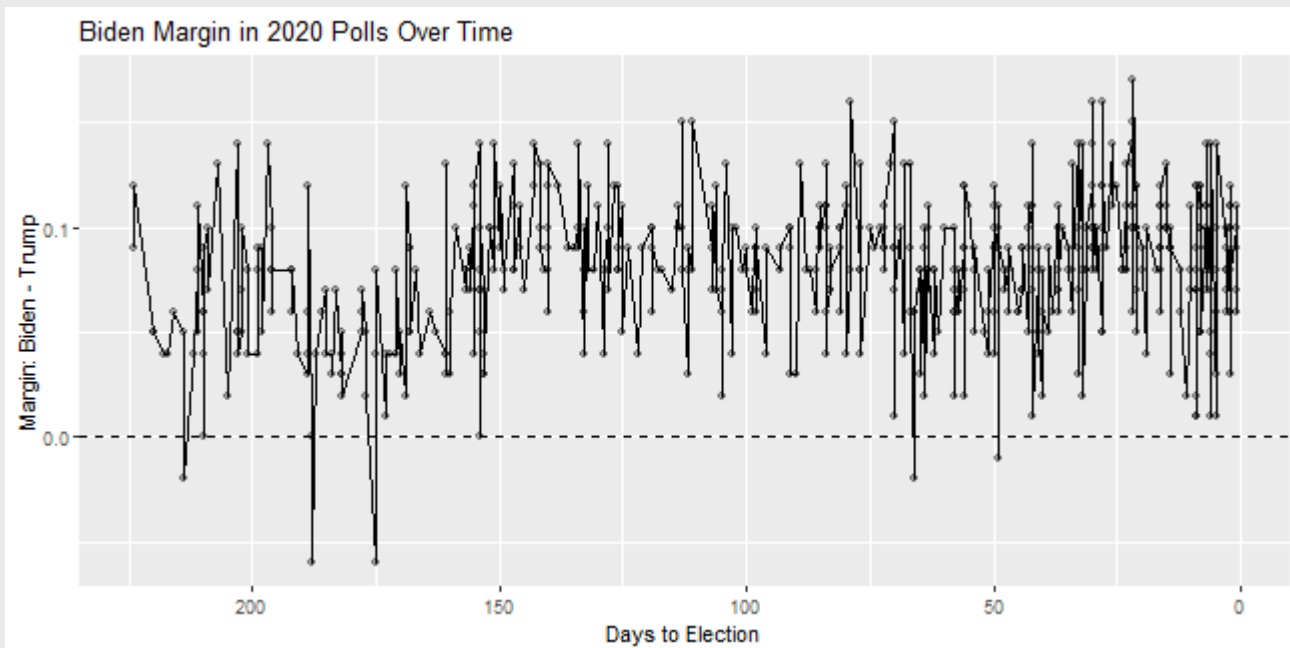
Margin by Time to Election

```
toplot %>%  
  ggplot(aes(x = DaysToED,y = margin)) +  
  geom_jitter(height = .005) +  
  labs(title = 'Biden Margin in 2020 Polls Over Time',  
       x = 'Days to Election',y = 'Margin: Biden - Trump') +  
  scale_x_reverse() + geom_hline(yintercept = 0,linetype = 'dashed')
```



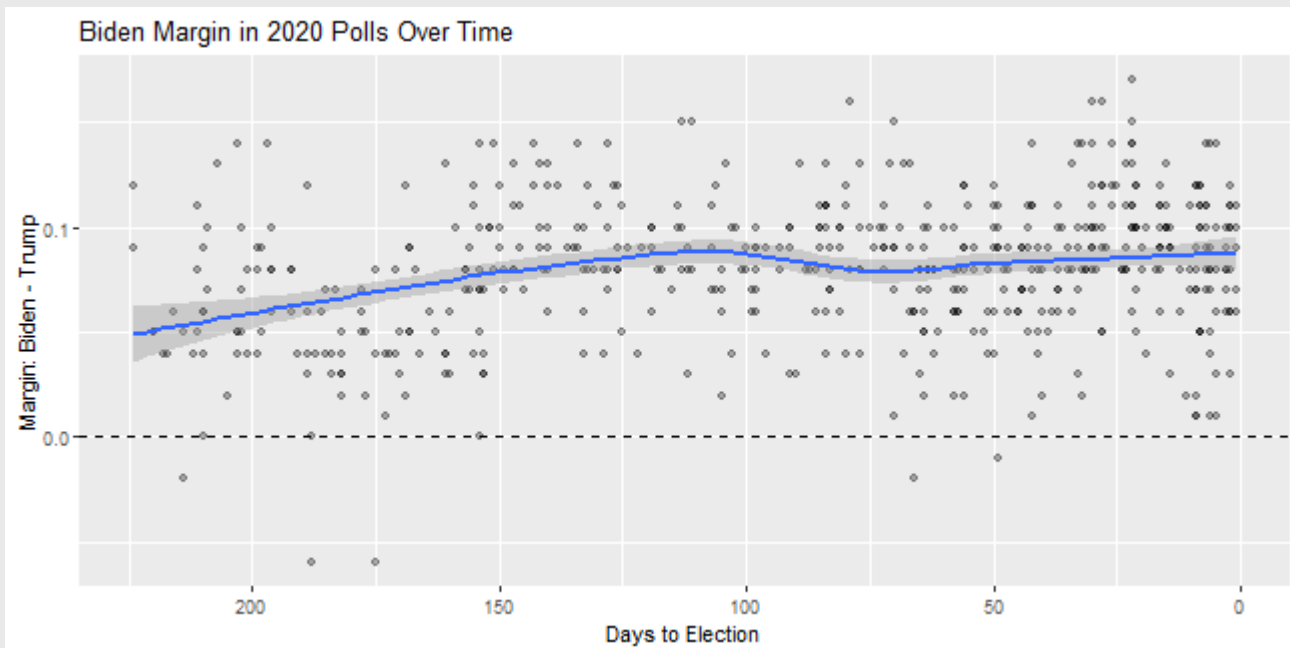
Highlight Trends

```
toplot %>%  
  ggplot(aes(x = DaysToED,y = margin)) +  
  geom_point(alpha = .3) +  
  geom_line() +  
  labs(title = 'Biden Margin in 2020 Polls Over Time',  
        x = 'Days to Election',y = 'Margin: Biden - Trump') +  
  scale_x_reverse() + geom_hline(yintercept = 0,linetype = 'dashed')
```



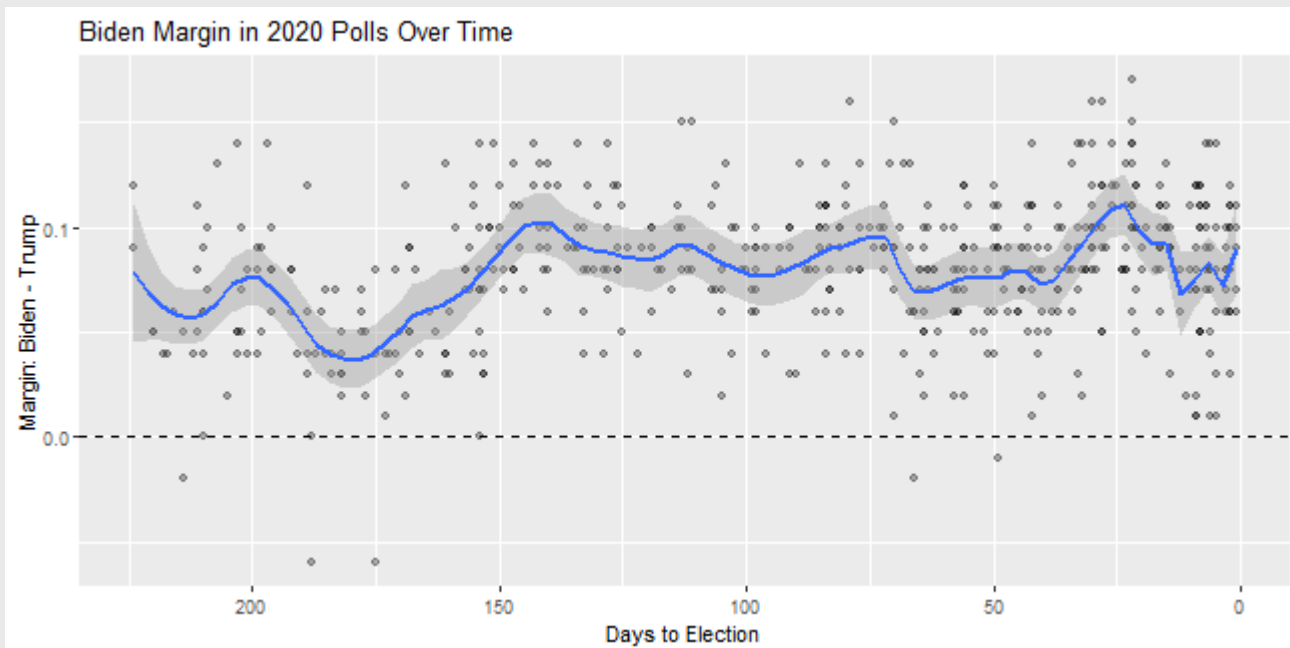
Highlight Trends

```
toplot %>%  
  ggplot(aes(x = DaysToED,y = margin)) +  
  geom_point(alpha = .3) +  
  geom_smooth() +  
  labs(title = 'Biden Margin in 2020 Polls Over Time',  
        x = 'Days to Election',y = 'Margin: Biden - Trump') +  
  scale_x_reverse() + geom_hline(yintercept = 0,linetype = 'dashed')
```



Highlight Trends

```
toplot %>%  
  ggplot(aes(x = DaysToED,y = margin)) +  
  geom_point(alpha = .3) +  
  geom_smooth(span = .1) +  
  labs(title = 'Biden Margin in 2020 Polls Over Time',  
        x = 'Days to Election',y = 'Margin: Biden - Trump') +  
  scale_x_reverse() + geom_hline(yintercept = 0,linetype = 'dashed')
```



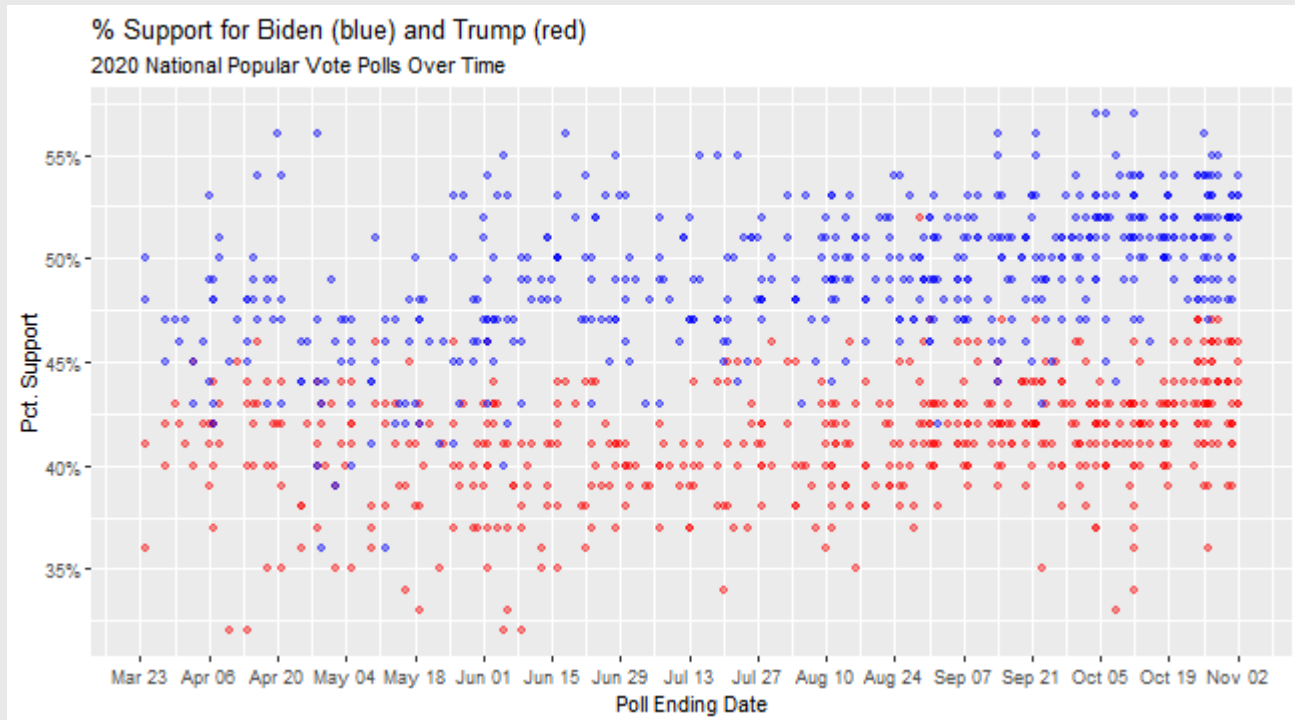
Multiple Variables over time

- We can look at multiple continuous variables at the same time
 - Use `color` or `fill` to distinguish

```
BidenTrumpplot <- topplot %>%  
  ggplot() +  
  geom_point(aes(x = EndDate, y = Trump),  
             color = "red", alpha=.4) +  
  geom_point(aes(x = EndDate, y = Biden),  
             color = "blue", alpha=.4) +  
  labs(title="% Support for Biden (blue) and Trump (red)",  
        subtitle = "2020 National Popular Vote Polls Over Time",  
        y = "Pct. Support",  
        x = "Poll Ending Date") +  
  scale_x_date(date_breaks = "2 week", date_labels = "%b %d") +  
  scale_y_continuous(breaks=seq(.3,.7,by=.05),  
                    labels= scales::percent_format(accuracy = 1))
```

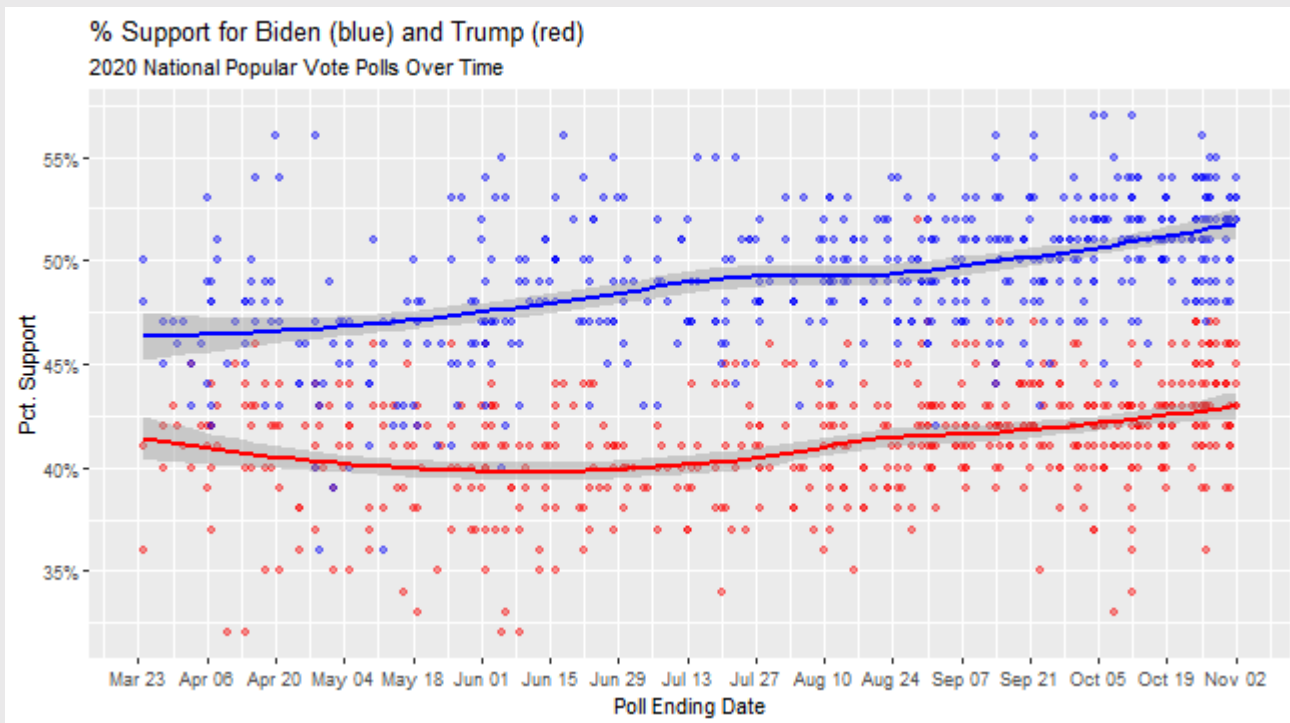
Multiple Variables over time

BidenTrumpplot



Highlighting Trends

```
BidenTrumpplot +  
  geom_smooth(aes(x = EndDate, y = Trump),  
              color = "red", se=T) +  
  geom_smooth(aes(x = EndDate, y = Biden),  
              color = "blue", se=T)
```



Typing back to theory

- Evidence of **both** Biden and Trump support increasing toward election
- Does this mean third party support waned?
- Does this mean that undecided voters made a decision?

2. Overtime Errors

- Wrangling to convert character dates to `date` class
- Plotting prediction errors over time

```
toplot <- poll %>%  
  mutate(EndDate = as.Date(EndDate, "%m/%d/%Y"),  
         StartDate = as.Date(StartDate, "%m/%d/%Y"),  
         DaysToED = as.numeric(as.Date('11/3/2020', format =  
"%m/%d/%Y") - EndDate),  
         margin = Biden - Trump,  
         errDem = Biden - DemCertVote / 100,  
         errRep = Trump - RepCertVote / 100)
```

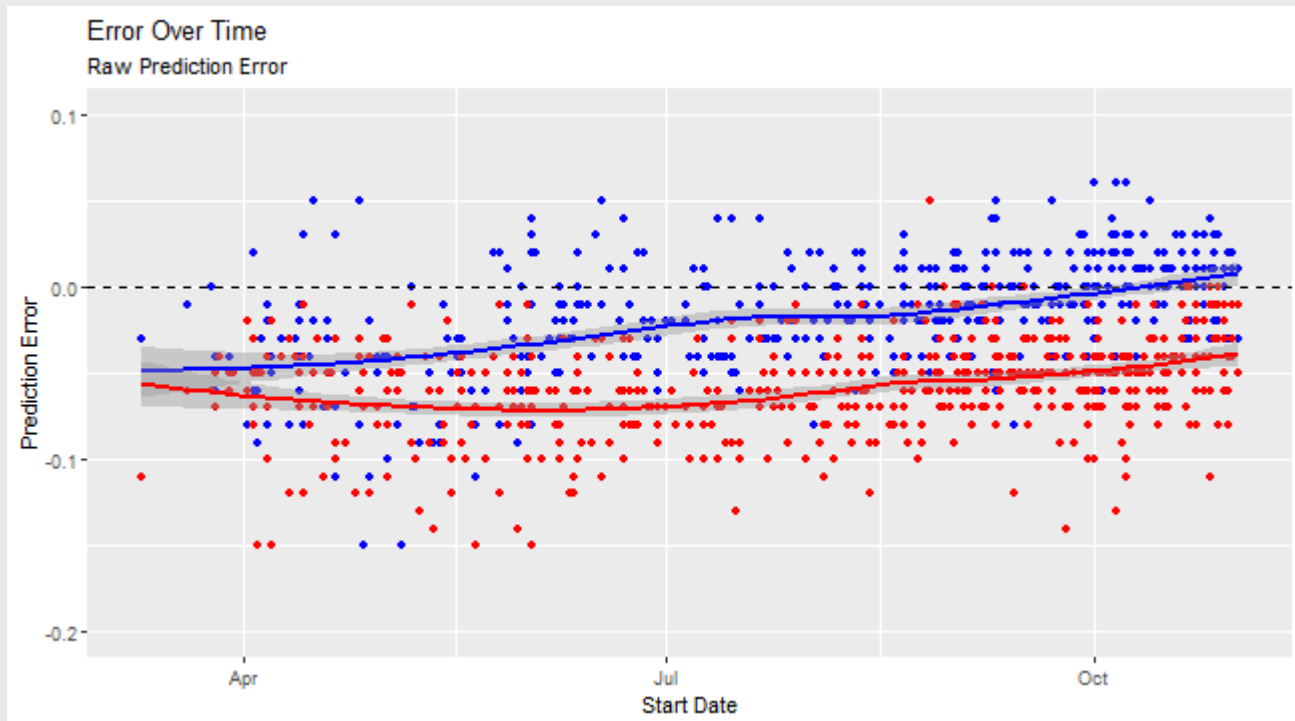
Overtime Errors

- Wrangling to convert character dates to `date` class
- Plotting prediction errors over time

```
pErrOverTime_Raw <- topplot %>%  
  ggplot(aes(x = StartDate)) +  
  geom_point(aes(y = errDem),color = 'blue') +  
  geom_point(aes(y = errRep),color = 'red') +  
  geom_hline(yintercept = 0,linetype = 'dashed') +  
  geom_smooth(aes(y = errDem),color = 'blue') +  
  geom_smooth(aes(y = errRep),color = 'red') +  
  labs(title = 'Error Over Time',subtitle = 'Raw Prediction Error',x  
= 'Start Date',y = 'Prediction Error') +  
  ylim(c(-.2,.1))
```

Overtime Errors

pErrOverTime_Raw



Poll Quality

- Polls got better
 - ...but more for Biden than for Trump
- Poll quality:
 - Sample size
 - "Margin of Error" (MoE)

Weighting by Sample Size

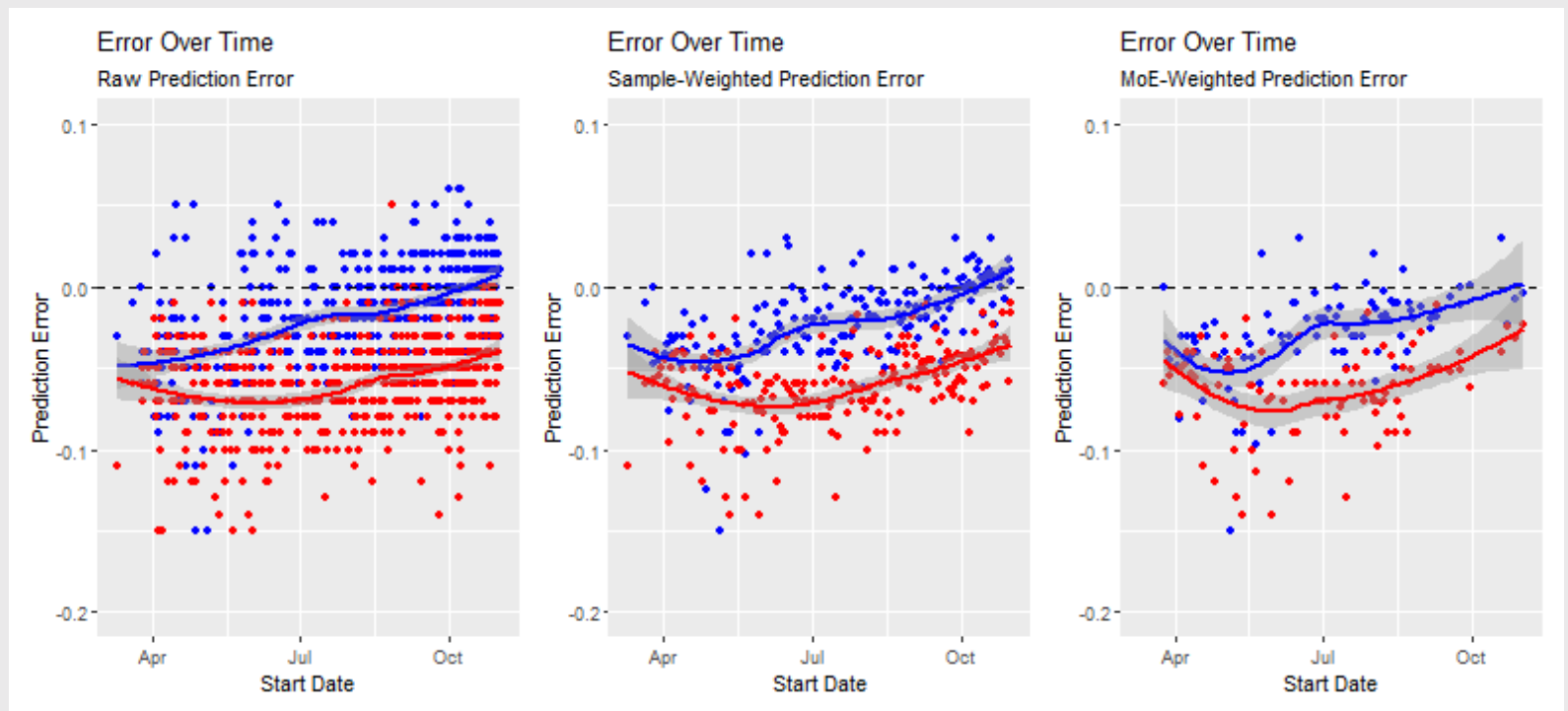
```
pErrOverTime_SampleSize <- topplot %>%  
  group_by(StartDate) %>%  
  summarise(errDem = weighted.mean(errDem,w = SampleSize),  
            errRep = weighted.mean(errRep,w = SampleSize)) %>%  
  ggplot(aes(x = StartDate)) +  
  geom_point(aes(y = errDem),color = 'blue') +  
  geom_point(aes(y = errRep),color = 'red') +  
  geom_hline(yintercept = 0,linetype = 'dashed') +  
  geom_smooth(aes(y = errDem),color = 'blue') +  
  geom_smooth(aes(y = errRep),color = 'red') +  
  labs(title = 'Error Over Time',subtitle = 'Sample-Weighted  
Prediction Error',x = 'Start Date',y = 'Prediction Error') +  
  ylim(c(-.2,.1))
```

Weighting by Inverse of MoE

```
pErrOverTime_MoE <- toplot %>%  
  group_by(StartDate) %>%  
  summarise(errDem = weighted.mean(errDem,w = 1/MoE),  
            errRep = weighted.mean(errRep,w = 1/MoE)) %>%  
  ggplot(aes(x = StartDate)) +  
  geom_point(aes(y = errDem),color = 'blue') +  
  geom_point(aes(y = errRep),color = 'red') +  
  geom_hline(yintercept = 0,linetype = 'dashed') +  
  geom_smooth(aes(y = errDem),color = 'blue') +  
  geom_smooth(aes(y = errRep),color = 'red') +  
  labs(title = 'Error Over Time',subtitle = 'MoE-Weighted Prediction  
Error',x = 'Start Date',y = 'Prediction Error') +  
  ylim(c(-.2,.1))
```

Comparing

```
require(patchwork)
pErrOverTime_Raw + pErrOverTime_SampleSize + pErrOverTime_MoE
```



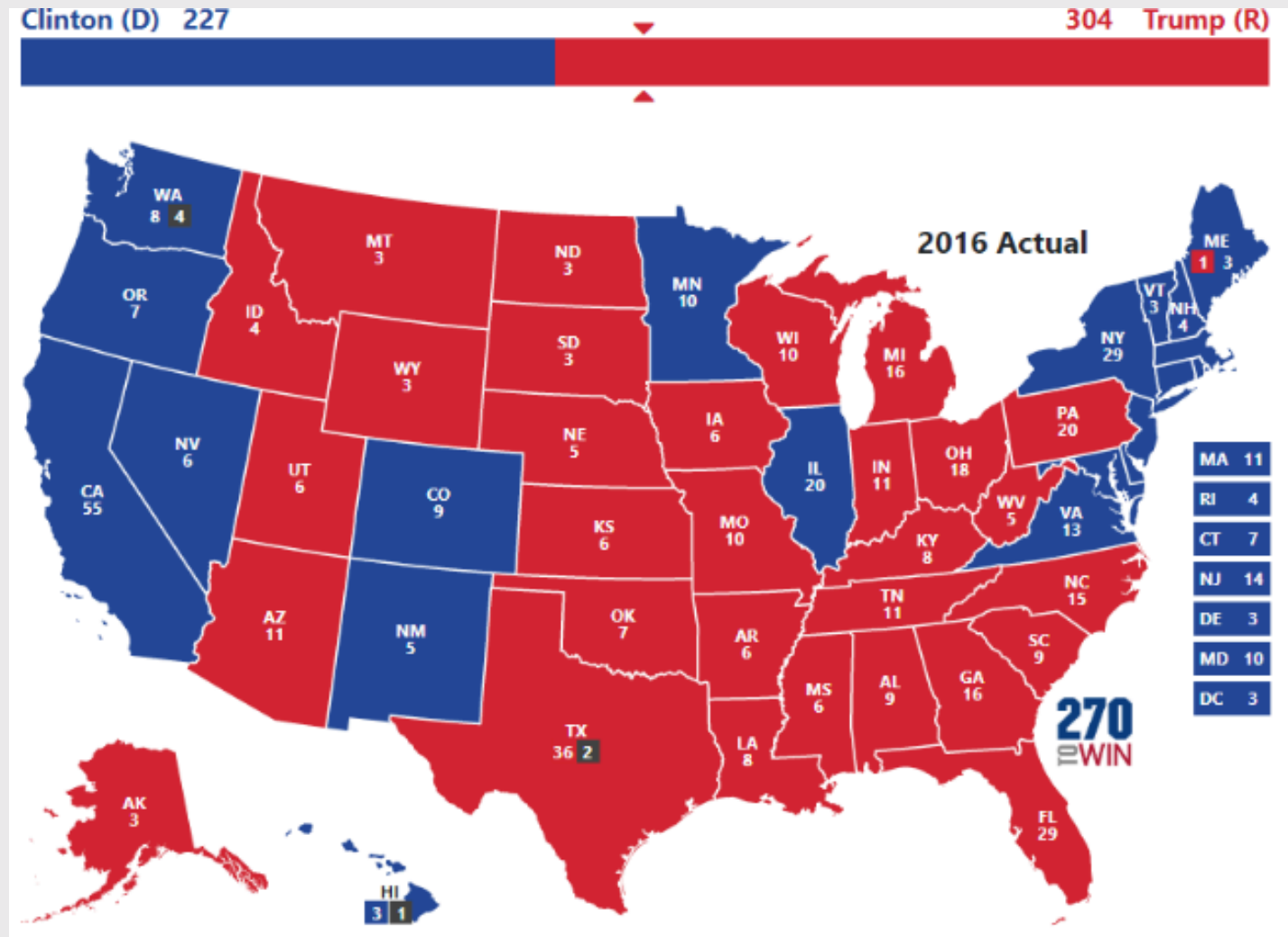
3. The Electoral College

- U.S. presidents determined by the **Electoral College** (EC)
 - Each state allocated votes equal to their "congressional delegation"
 - One vote per Senator + one vote per **Member of the House** (MoH)
 - 100 Senators + 435 MoHs + 3 votes for DC = 538 total votes
 - To win the presidency, you need at least 270 EC votes
- What does this mean for representation?
 - Population divided equally* across 435 MoHs
 - In 2020, each MoH represented 761,179 people
 - Geographic distribution of this population is **non-uniform**

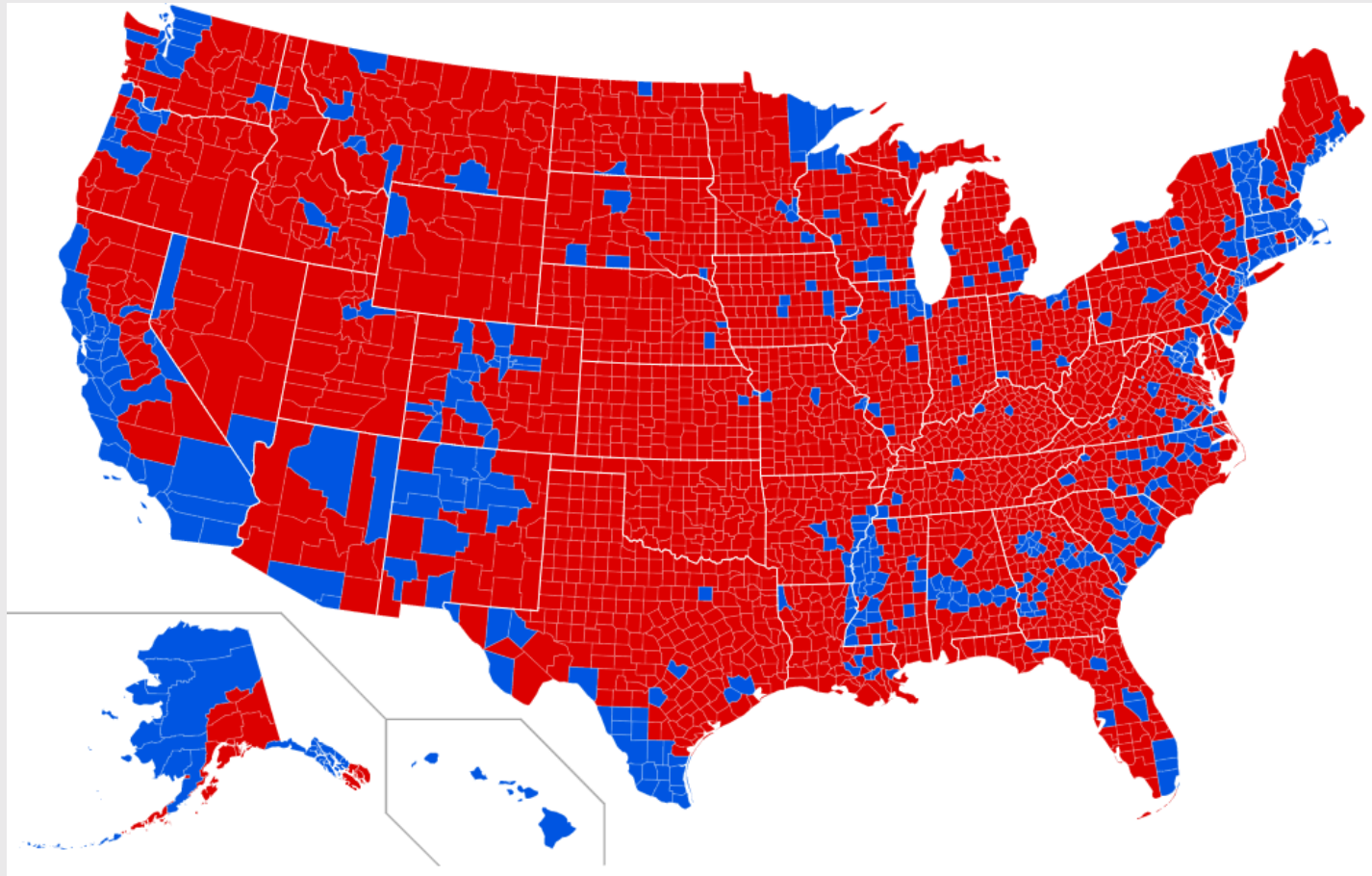
Assigning Districts

- Every 10 years, U.S. takes a census of its population
- Based on these results, congressional districts are reapportioned
- In 2010, Florida's pop. was ~19m & average pop. per MoH was ~710k
 - How many MoHs? $\text{ceiling}(19000000 / 710000) = 27$
 - Every state has 2 senators, so FL had $27 + 2 = 29$ EC votes
 - In 2020, FL pop. was 21m → each resident had 0.00000137 of a vote
- What about Vermont?
 - $3 \text{ EC votes} / 624,340 \text{ residents} = 0.00000481$
- Vermonters have roughly 3.5 times the influence of Floridians!

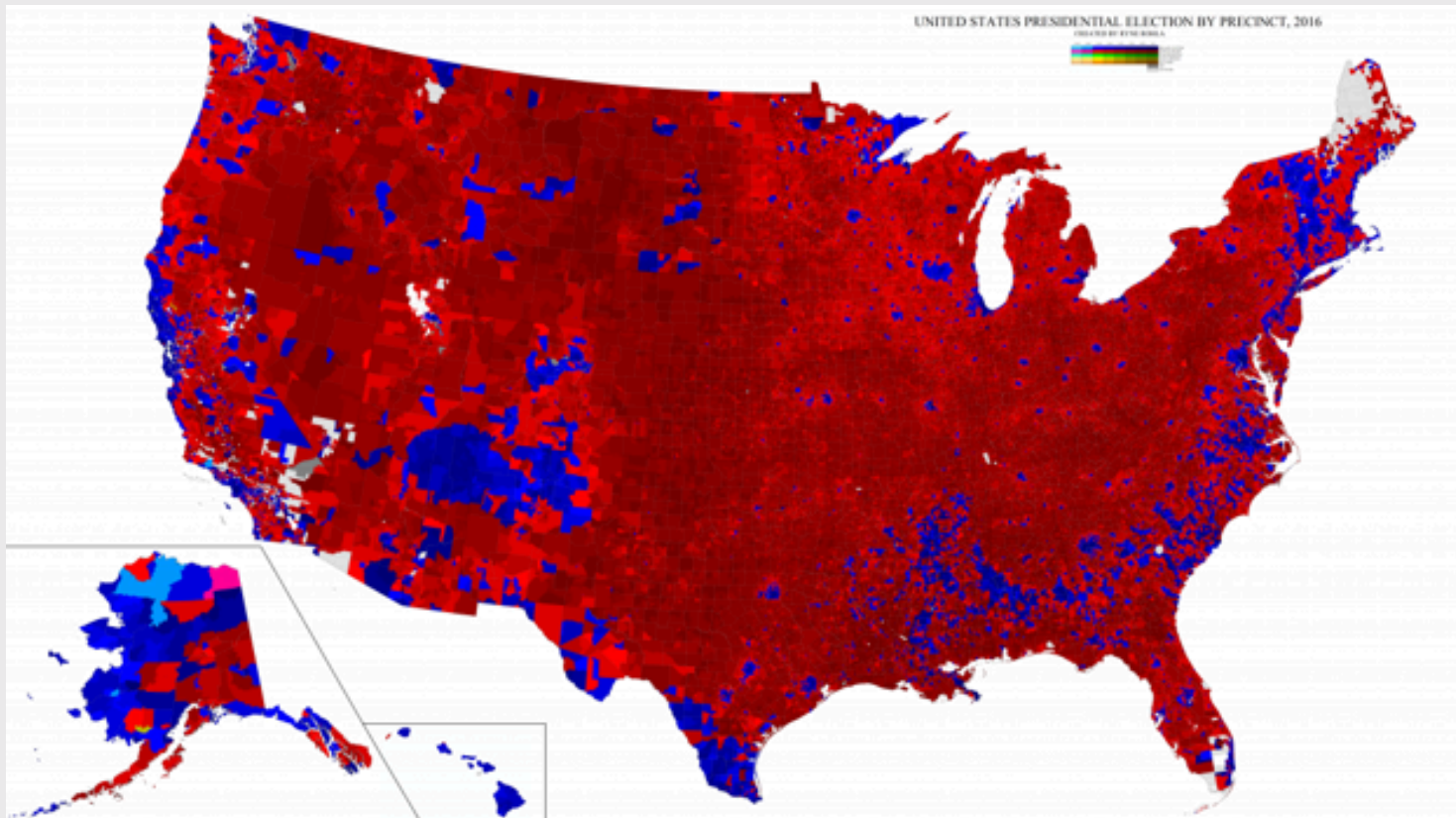
Data Viz Misleads!



Data Viz Misleads!



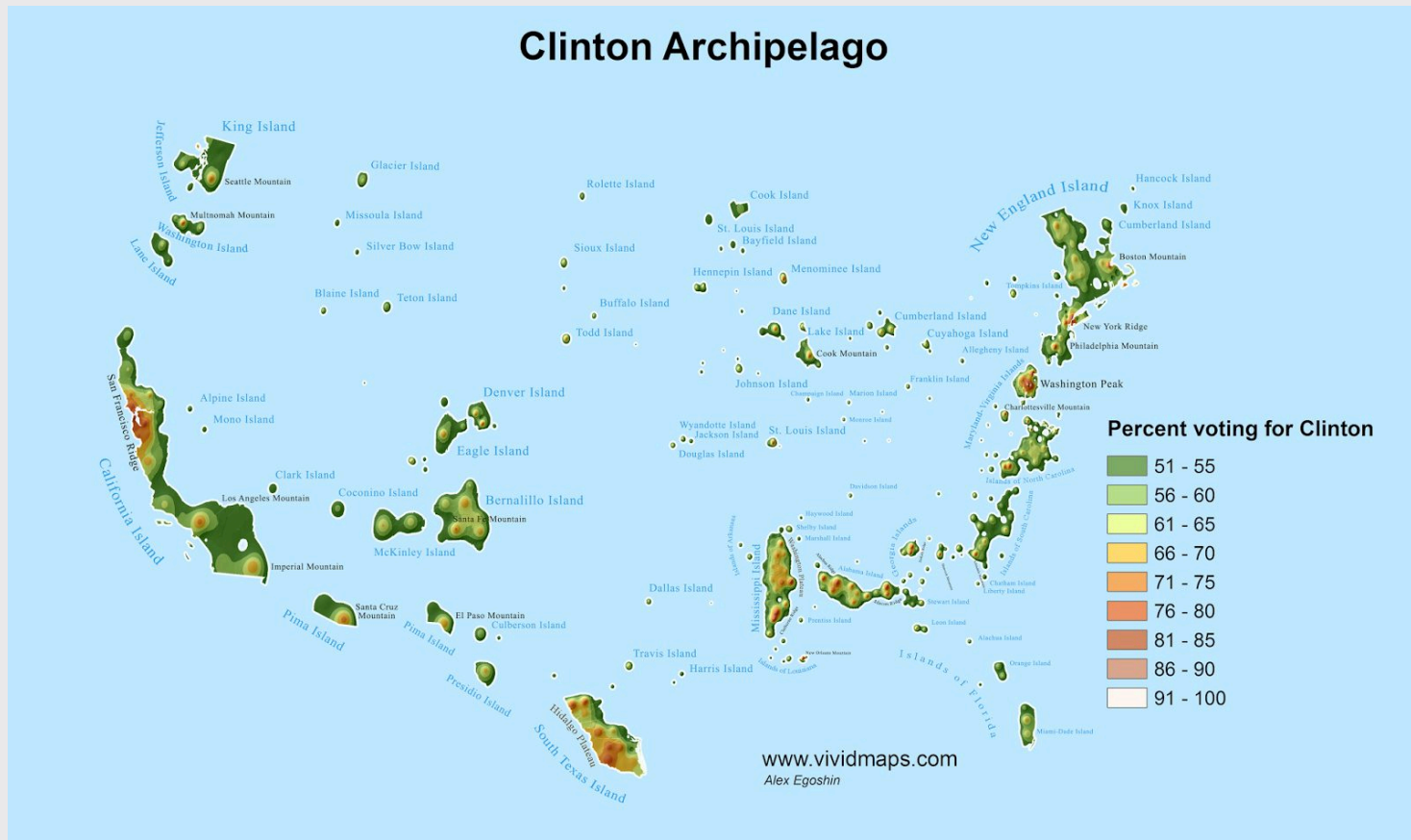
Data Viz Misleads!



Data Viz Misleads!

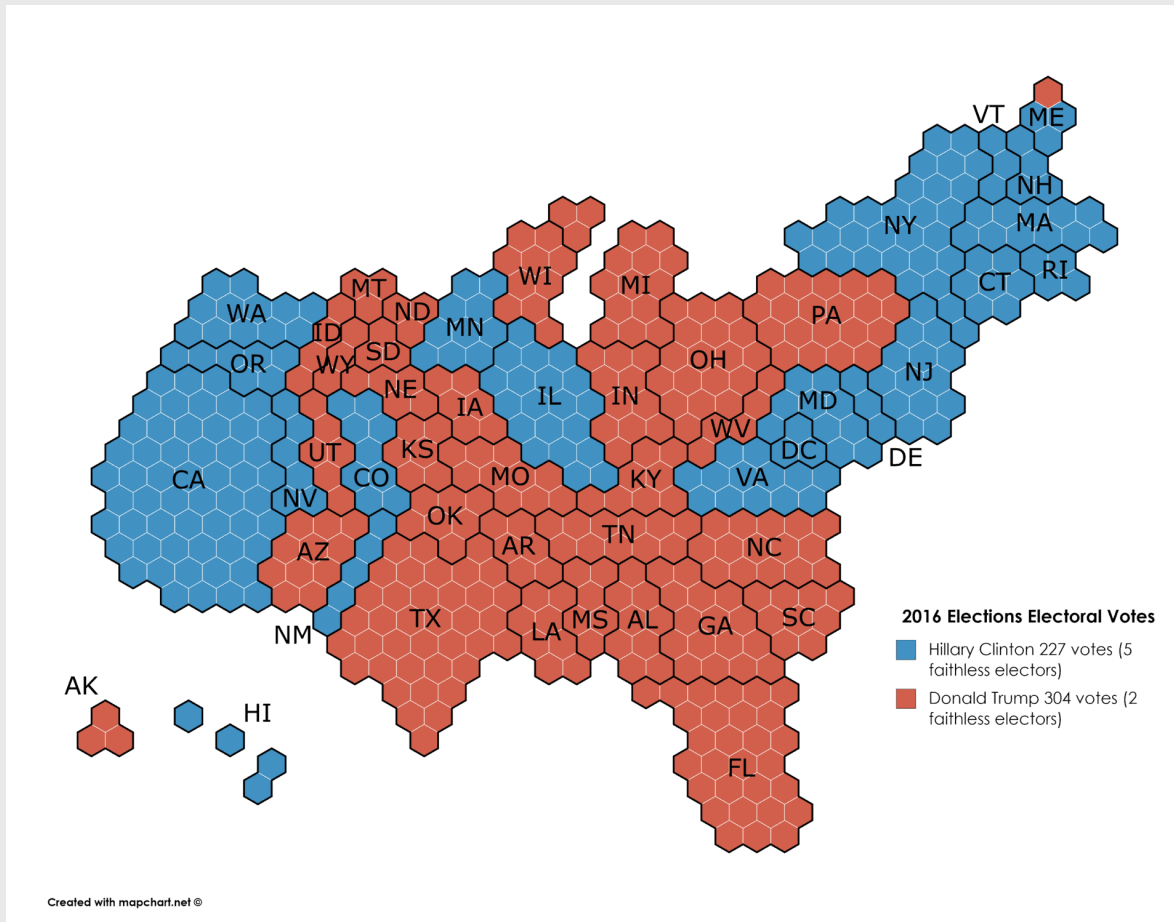


Data Viz Misleads!



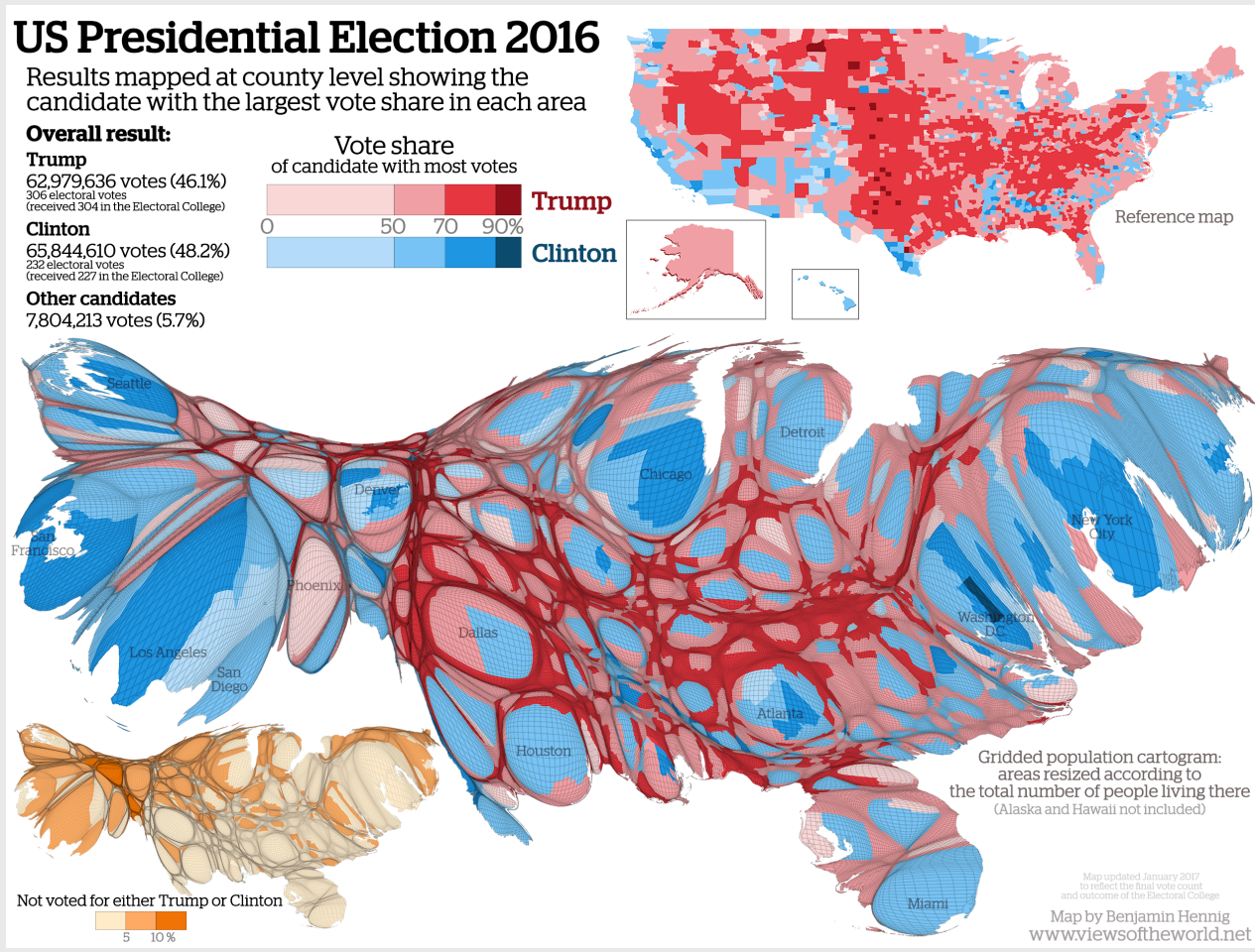
Correcting Maps

- Fundamental challenge due to distribution of population across land



Correcting Maps

- Fundamental challenge due to distribution of population across land



Introducing Data

- We will attempt to use state polls to predict 2020 winner
- National polls are great at predicting popular vote share...
- ...but presidents aren't elected based on popular vote
- With a few exceptions, states have a "winner take all" system
 - Winner of state's popular vote gets **all** of the state's EC votes
- Thus we need good state-level polls
- Load `Pres2020_StatePolls.Rds` to `statePoll` object

```
require(tidyverse)
statePoll <-
read_rds('https://github.com/jbisbee1/DS1000_F2024/raw/main/data/Pres20
```

Look at the data

```
glimpse(statePoll)
```

```
## Rows: 1,545
## Columns: 19
## $ StartDate      <date> 2020-03-21, 2020-03-24, 2020-03-24...
## $ EndDate        <date> 2020-03-30, 2020-04-03, 2020-03-29...
## $ DaysinField    <dbl> 10, 11, 6, 2, 3, 5, 2, 2, 7, 3, 3, ...
## $ MoE            <dbl> 2.8, 3.0, 4.2, NA, 4.0, 1.7, 3.0, 3...
## $ Mode           <chr> "Phone/Online", "Phone/Online", "Li...
## $ SampleSize     <dbl> 1331, 1000, 813, 962, 602, 3244, 10...
## $ Biden          <dbl> 41, 47, 48, 67, 46, 46, 46, 48, 52,...
## $ Trump          <dbl> 46, 34, 45, 29, 46, 40, 48, 45, 39,...
## $ Winner         <chr> "Rep", "Dem", "Dem", "Dem", "Dem", ...
## $ poll.predicted <dbl> 1, 1, 1, 1, 0, 0, 0, 1, 1, 1, 1, 1,...
## $ Funded         <chr> "UtahPolicy.com & KUTV 2News", "Sac...
## $ Conducted      <chr> "Y2 Analytics", "GreatBlue Research...
## $ margin         <dbl> -5, 13, 3, 38, 0, 6, -2, 3, 13, -7,...
## $ DaysToED       <drtn> 218 days, 214 days, 219 days, 219 ...
## $ StateName      <chr> "Utah", "Connecticut", "Wisconsin",...
## $ EV             <int> 6, 7, 10, 55, 16, 29, 16, 16, 12, 1...
## $ State          <chr> "UT", "CT", "WI", "CA", "MI", "FL",...
## $ BidenCertVote  <dbl> 38, 59, 49, 64, 51, 48, 50, 51, 58,...
```


Scientific Goal: Predict Winner

- Need to calculate predicted probability of Biden victory
 1. Fraction of polls with Biden in the lead
 2. Fraction of respondents supporting Biden
 3. Fraction of twoway respondents supporting Biden
- Wrangling time!

```
statePoll <- statePoll %>%  
  mutate(Biden2W = Biden / (Biden+Trump),  
         Biden = Biden / 100,  
         Trump = Trump/100)  
  
stateProbs <- statePoll %>%  
  group_by(StateName,State) %>%  
  summarise(BidenProb1 = mean(Biden > Trump),  
            BidenProb2 = mean(Biden),  
            BidenProb3 = mean(Biden2W),  
            EV = mean(EV),.groups = 'drop') # This is just the value
```

of EV

Looking at data

```
stateProbs
```

```
## # A tibble: 50 × 6
##   StateName State BidenProb1 BidenProb2 BidenProb3 EV
##   <chr>      <chr>      <dbl>      <dbl>      <dbl> <dbl>
## 1 Alabama    AL          0          0.389      0.407    9
## 2 Alaska     AK          0          0.442      0.466    3
## 3 Arizona    AZ          0.840      0.484      0.519   11
## 4 Arkansas   AR          0          0.381      0.395    6
## 5 California CA          1          0.618      0.661   55
## 6 Colorado   CO          1          0.534      0.571    9
## 7 Connecticut CT          1          0.584      0.631    7
## 8 Delaware   DE          1          0.603      0.627    3
## 9 Florida    FL          0.798      0.486      0.517   29
## 10 Georgia   GA          0.548      0.474      0.504   16
## # i 40 more rows
```


Visualizing Data

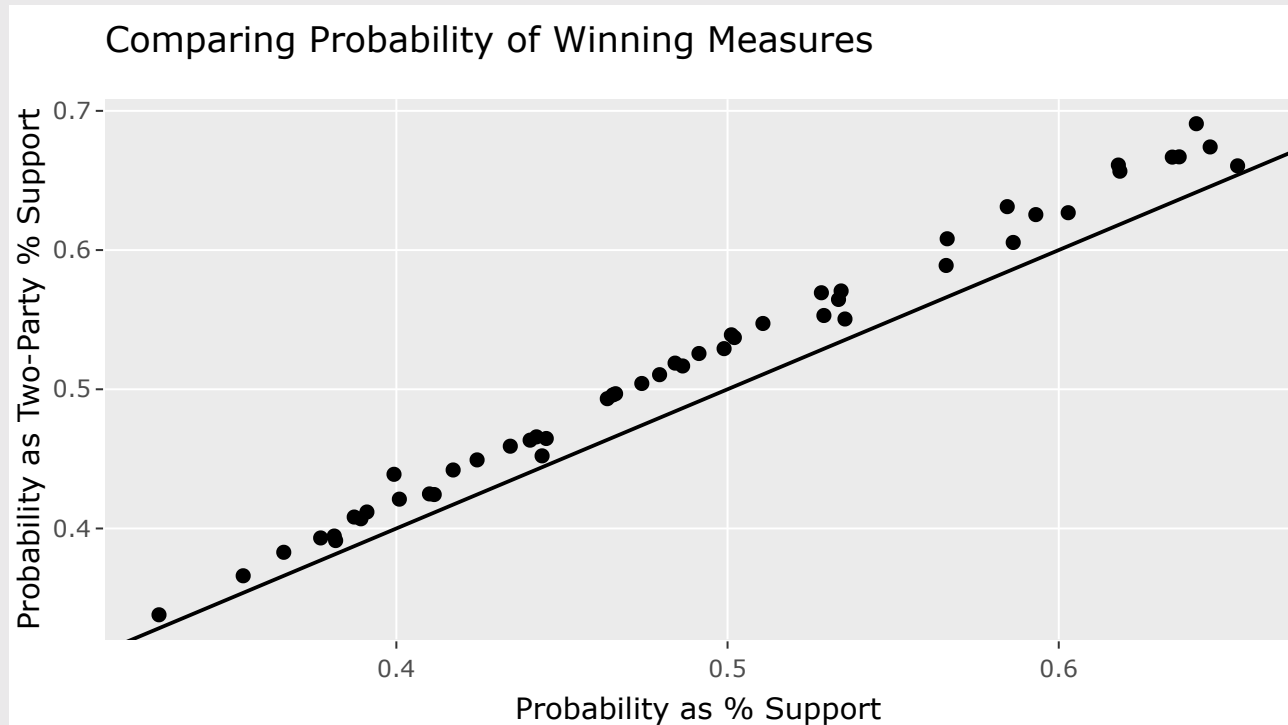
- R has fancy packages that allow for interactive plots (`plotly`)

```
# install.packages('plotly')
require(plotly)

gg <- stateProbs %>%
  ggplot(aes(x=BidenProb2, y=BidenProb3, text=paste(StateName))) +
  geom_point() +
  geom_abline(intercept=0, slope=1) +
  labs(x= "Probability as % Support",
       y = "Probability as Two-Party % Support",
       title = "Comparing Probability of Winning Measures")
```

Visualizing Data

```
ggplotly(gg, tooltip = 'text')
```



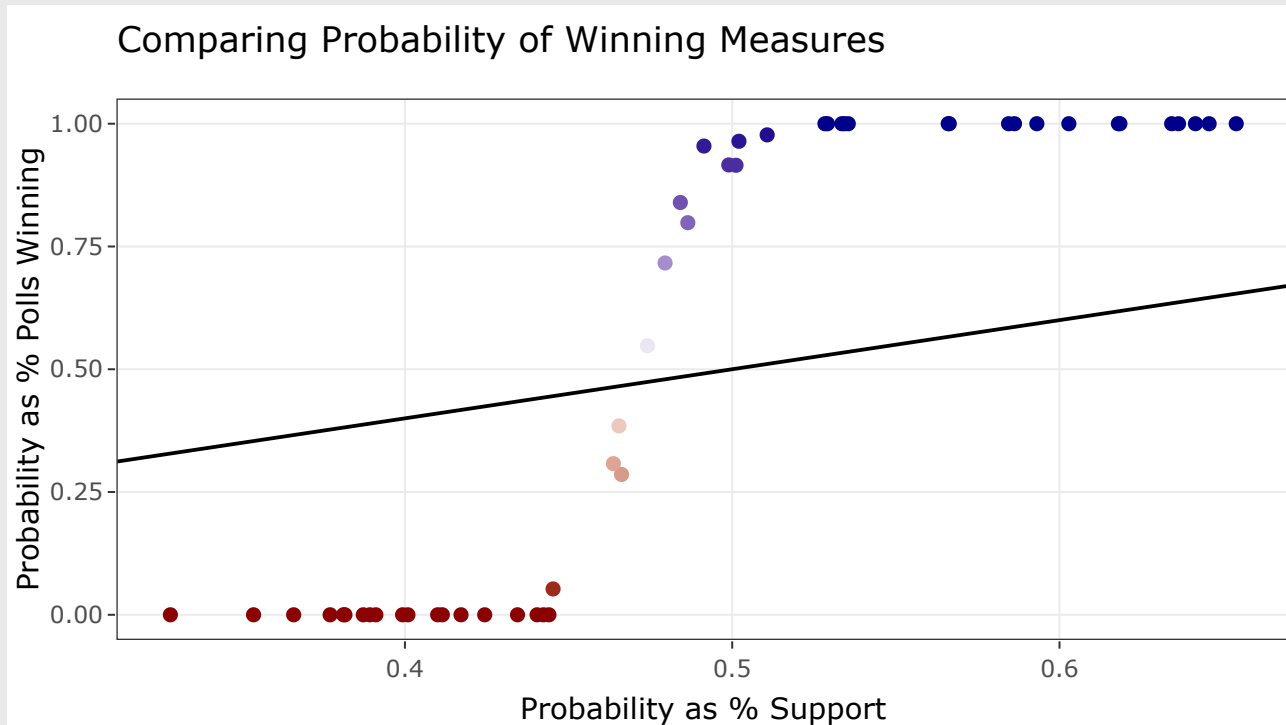
Visualizing Data

- What about the % of polls that predict a Biden victory?
 - I.e., we know Trump won't win California or Biden won't win Tennessee

```
gg <- stateProbs %>%  
  ggplot(aes(x=BidenProb2, y=BidenProb1, text=paste(StateName), color =  
BidenProb1)) +  
  geom_point() +  
  geom_abline(intercept=0, slope=1) +  
  labs(x= "Probability as % Support",  
       y = "Probability as % Polls Winning",  
       title = "Comparing Probability of Winning Measures") +  
  scale_color_gradient2(guide = 'none', low = "darkred", mid =  
"white", high = "darkblue", midpoint = .5) +  
  theme_bw()
```

Visualizing Data

```
ggplotly(gg,tooltip = 'text')
```



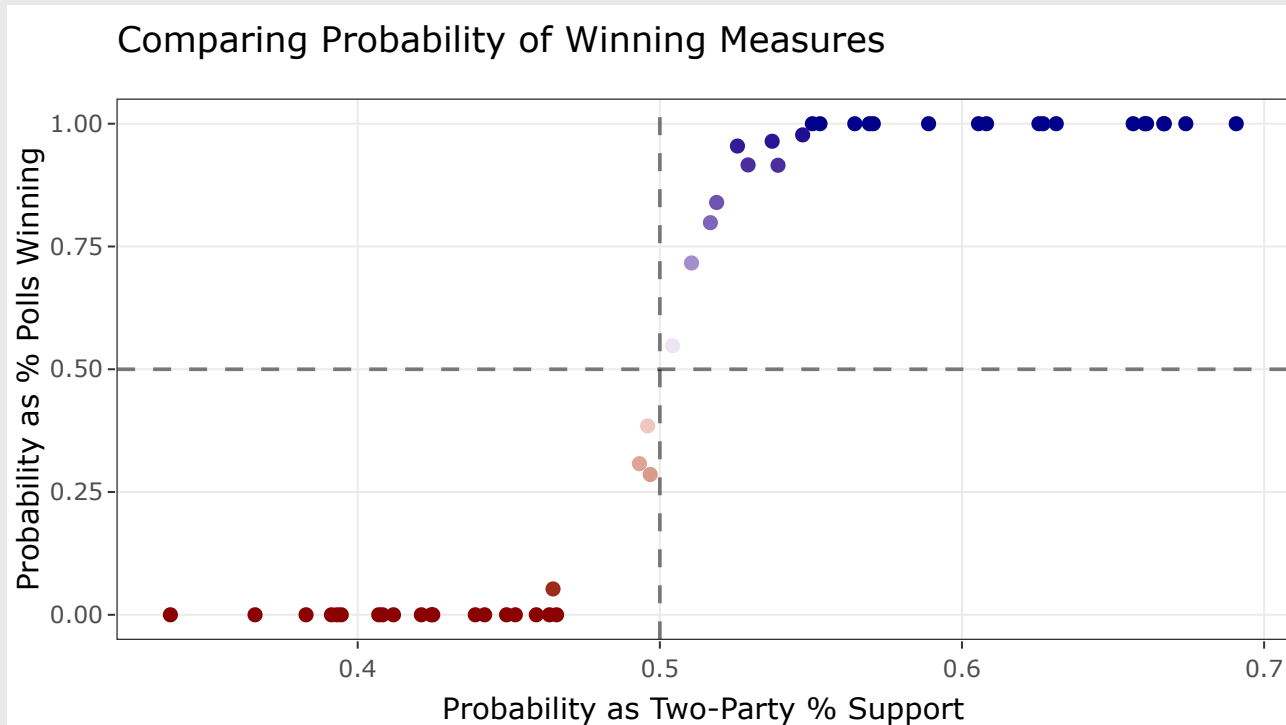
Visualizing Data

- Comparing to two-way

```
gg <- stateProbs %>%  
  ggplot(aes(x=BidenProb3, y=BidenProb1, text=paste(StateName), color =  
BidenProb1)) +  
  geom_point() +  
  labs(x= "Probability as Two-Party % Support",  
       y = "Probability as % Polls Winning",  
       title = "Comparing Probability of Winning Measures") +  
  scale_color_gradient2(guide = 'none', low = "darkred", mid =  
"white", high = "darkblue", midpoint = .5) +  
  theme_bw() +  
  geom_hline(yintercept = .5, linetype = 'dashed', alpha = .5) +  
  geom_vline(xintercept = .5, linetype = 'dashed', alpha = .5)
```

Visualizing Data

```
ggplotly(gg,tooltip = 'text')
```



Translate to EC Votes

- Start with a single state to get the idea
- **Expected Value** (EV...could also be Expected Votes): Probability * Value

```
stateProbs %>%  
  filter(State == 'PA') %>%  
  mutate(BEV1 = BidenProb1 * EV,  
         BEV2 = BidenProb2 * EV,  
         BEV3 = BidenProb3 * EV) %>%  
  select(EV, matches('^B.*\\d'))
```

```
## # A tibble: 1 × 7  
##       EV BidenProb1 BidenProb2 BidenProb3 BEV1 BEV2 BEV3  
##   <dbl>      <dbl>      <dbl>      <dbl> <dbl> <dbl> <dbl>  
## 1     20      0.916      0.499      0.529  18.3  9.98  10.6
```

- Probability matters! Biden could get between 18.3 and 9.98 votes in PA!

Translate to EC Votes

- Do this for every state

```
stateProbs %>%  
  mutate(BidenEV1 = BidenProb1 * EV,  
         BidenEV2 = BidenProb2 * EV,  
         BidenEV3 = BidenProb3 * EV,  
         TrumpEV1 = EV - BidenEV1,  
         TrumpEV2 = EV - BidenEV2,  
         TrumpEV3 = EV - BidenEV3) %>%  
  summarise_at(vars(matches('EV\\d')),sum) # Fancy shortcut!
```

```
## # A tibble: 1 × 6  
##   BidenEV1 BidenEV2 BidenEV3 TrumpEV1 TrumpEV2 TrumpEV3  
##   <dbl>    <dbl>    <dbl>    <dbl>    <dbl>    <dbl>  
## 1     345.     272.     289.     190.     263.     246.
```

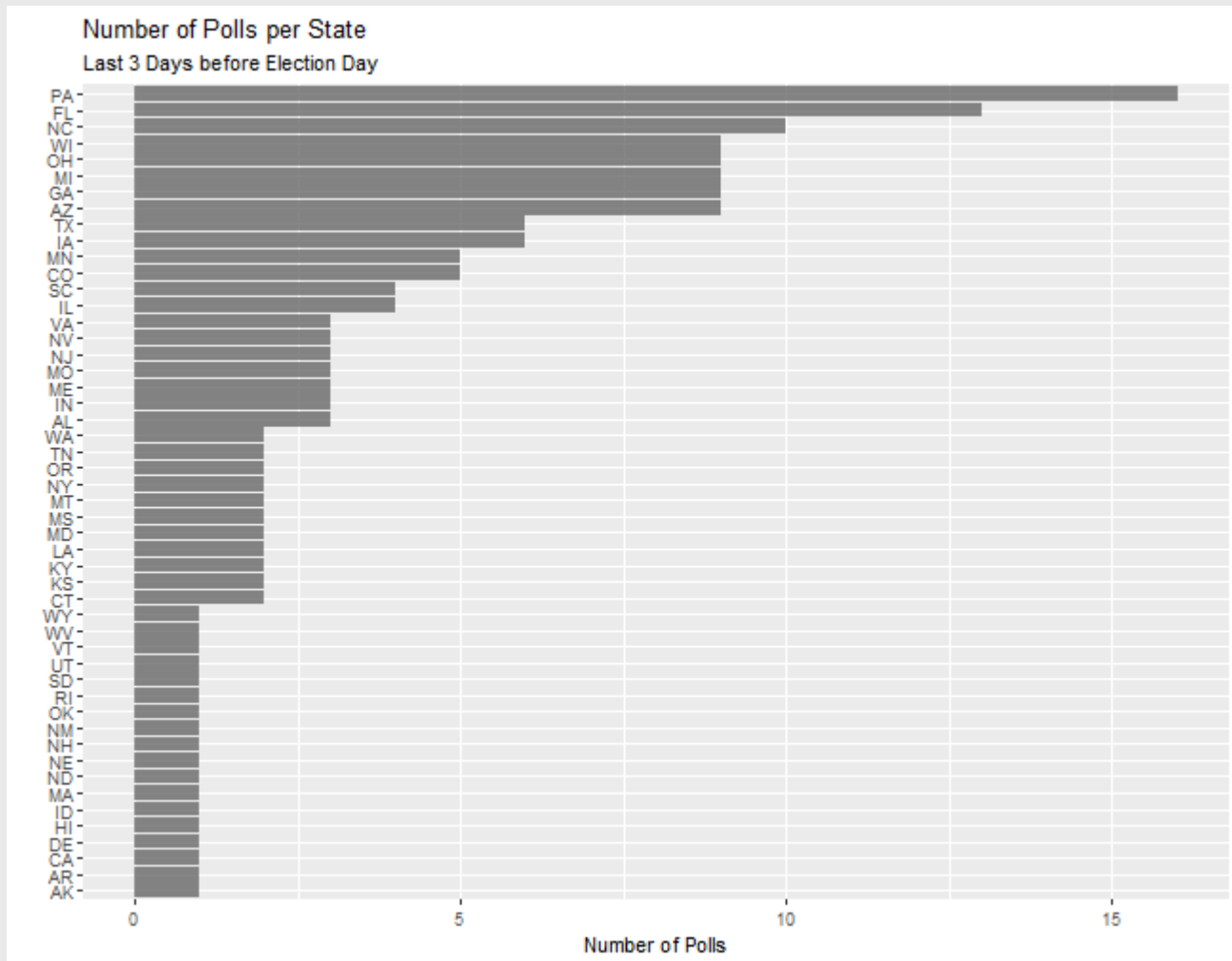
- Overall EC votes *a/ways* favored Biden (always above 270)
- But none of our estimates are close to the true outcome of 306 to 232

Science Questions

- We know that **national** polls changed closer to election
- Can we do better if we look only at **state** polls close to election day?
- **Data** limitations!

```
pollsPerState <- statePoll %>%  
  filter(DaysToED <= 3) %>%  
  count(State) %>%  
  ggplot(aes(x = n, y = reorder(State, n))) +  
  geom_bar(stat = 'identity', alpha = .7) +  
  labs(title = 'Number of Polls per State',  
        subtitle = 'Last 3 Days before Election Day',  
        x = 'Number of Polls',  
        y = '')
```

Data Limitations



Do better with recent polls?

```
statePoll %>%  
  filter(DaysToED <= 3) %>%  
  group_by(StateName, State) %>%  
  summarise(BidenProb1 = mean(Biden > Trump),  
            BidenProb2 = mean(Biden),  
            BidenProb3 = mean(Biden2W),  
            EV = mean(EV), .groups = 'drop') %>%  
  mutate(BidenEV1 = BidenProb1 * EV,  
         BidenEV2 = BidenProb2 * EV,  
         BidenEV3 = BidenProb3 * EV,  
         TrumpEV1 = EV - BidenEV1,  
         TrumpEV2 = EV - BidenEV2,  
         TrumpEV3 = EV - BidenEV3) %>%  
  summarise_at(vars(matches('EV\\d')), sum)
```

```
## # A tibble: 1 × 6  
##   BidenEV1 BidenEV2 BidenEV3 TrumpEV1 TrumpEV2 TrumpEV3  
##   <dbl>    <dbl>    <dbl>    <dbl>    <dbl>    <dbl>  
## 1     334.     277.     285.     201.     258.     250.
```

Conclusion

- Still not nailing the true votes!
 - Not even within the distributions of the bootstrapped estimates!
- Why?
 1. Polls have their own uncertainty (could weight by total respondents?)
 2. Polls might not be representative of voters
 3. Polls might not be independent of each other

Quiz & Homework

- Go to Brightspace and take the **8th** quiz
 - The password to take the quiz is ####
- **Homework:**
 1. Work through ds1000_hw_9.Rmd
 2. Problem Set 5 (Brightspace)

